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COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

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Quiz #1

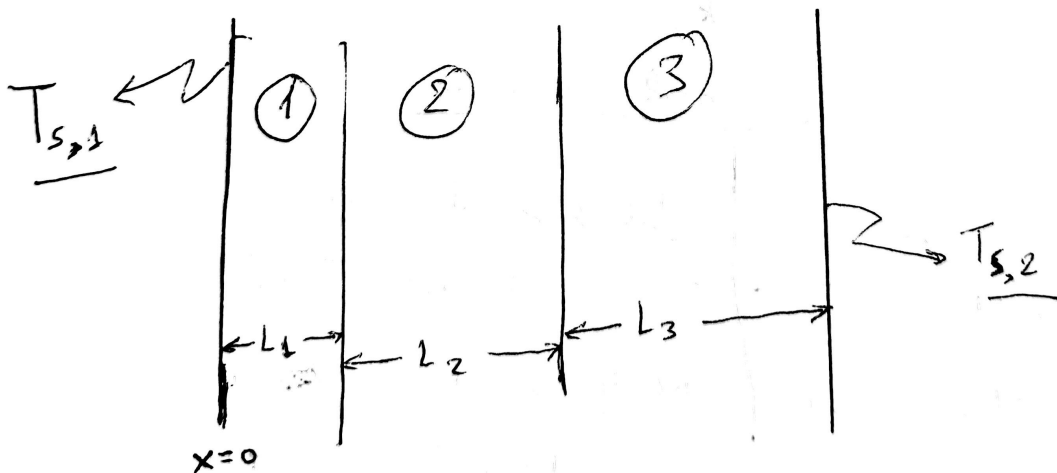
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Open Notes and Book

Duration: 40 minutes

Consider three sequential one-dimensional plane walls of thickness L_1 , L_2 and L_3 , respectively; heat is being transferred through this set of plane walls since the surface temperatures of the walls are maintained at $T_{s,1}$ and $T_{s,2}$ as shown in the sketch. The first and third ones experience $\dot{q} = \dot{q}_0(1 + a \cdot x)$ volumetric heat generation; the middle one involves uniform volumetric heat removal $\hat{q}$. All properties are constant.

- 1) Write all design/balance equations and corresponding boundary conditions
- 2) Determine the temperature profile $T(x)$ and heat flux distribution $q''(x)$.



② For each region

① ~~0 ≤ x ≤ L₁~~ $0 \leq x \leq L_1$
 conductivity k_1
 $q_1(x) = q_{01}(1 + a_1 x)$

② $L_1 \leq x \leq L_1 + L_2$
 conductivity k_2 , uniform removal hence $q_2(x) = \hat{q}$

③ $L_1 + L_2 \leq x \leq L_1 + L_2 + L_3$, conductivity k_3
 $q_3(x - (L_1 + L_2)) = q_{03}(1 + a_3 x)$

$k_1 \frac{d^2 T_1}{dx^2} = - \frac{q_1}{k_1} (1 + a_1 x)$ region ①

$k_2 \frac{d^2 T_2}{dx^2} = - \frac{\hat{q}}{k_2}$ region ②

$k_3 \frac{d^2 T_3}{dx^2} = - \frac{q_3}{k_3}$ region ③

Total 6 B.C.s

at $x=0$

BC1 $\rightarrow T_1(0) = T_{s,1}$

at $x = L_1 + L_2 + L_3$

BC2 $\rightarrow T(L_1 + L_2 + L_3) = T_{s,2}$

BC5 $\rightarrow (x = L_1 + L_2)$

$T_2(L_1 + L_2) = T_3(L_1 + L_2)$

BC3 $\rightarrow$ at $x = L_1$

$T_1(L_1) = T_2(L_1)$

BC4 $\rightarrow$ at $x = L_1$

$k_1 \frac{dT_1}{dx} \Big|_{x=L_1} = k_2 \frac{dT_2}{dx} \Big|_{x=L_1}$

BC6 at $x = L_1 + L_2$

$k_2 \frac{dT_2}{dx} \Big|_{x=L_1+L_2} = k_3 \frac{dT_3}{dx} \Big|_{x=L_1+L_2}$

Heat flux in each layer

Region 1 $\frac{dq}{dx} = -q_{01} (1 + a_1 x)$

(D)

$$q_{01}(x) = q_0 - q_{01} \left(x + \frac{a_1 x^2}{2} \right)$$

at $x = L_1$

$$q_{12} = q_{01}(L_1) = q_0 - q_{01} \left(L_1 + \frac{a_1 L_1^2}{2} \right)$$

Region 2

$$q_{03}(x) = q_{12} - q_{03} \left([x - (L_1 + L_2)] + \frac{a_3 [x - (L_1 + L_2)]^2}{2} \right)$$

let $x - (L_1 + L_2) = \alpha$

$$\therefore q_{03}(x) = q_{12} - q_{03} \left[\alpha + \frac{a_3 \alpha^2}{2} \right]$$

Temperature drops

Region 1 $\rightarrow \Delta T_1 = T(L_1^-) - T_{s,1} = \int_0^{L_1} \left(\frac{-q_{01}(x)}{k_1} \right) dx$

$$= -\frac{q_{01} L_1}{k_1} + \frac{q_{01}}{k_1} \left(\frac{L_1^2}{2} + \frac{a_1 L_1^3}{6} \right)$$

Region 2 $\rightarrow \Delta T_2 = T((L_1 + L_2)^-) - T(L_1^+)$

$$= \int_{L_1}^{L_1 + L_2} \left(\frac{-q_{02}(x)}{k_2} \right) dx = -\frac{q_{12} L_2}{k_2} - \frac{\hat{q} L_2^2}{2k_2}$$

Region 3 $\rightarrow \Delta T_3 = T_{s,2} - T((L_1 + L_2)^+) = \int_0^{L_3} \frac{-q_{23} - q_{03} \left(\alpha + \frac{a_3 \alpha^2}{2} \right)}{k_3} d\alpha$

$$= -\frac{q_{23} L_3}{k_3} + \frac{q_{03}}{k_3} \left(\frac{L_3^2}{2} + \frac{a_3 L_3^3}{6} \right)$$

$$T_{s,2} - T_{s,1} = \Delta T_1 + \Delta T_2 + \Delta T_3$$

$$\Delta T_1 = -\frac{q_{01} L_1}{k_1} + \frac{q_{01}}{k_1} \left(\frac{L_1^2}{2} + \frac{a L_1^3}{6} \right)$$

$$\Delta T_2 = -\frac{q_{12} L_2}{k_2} - \frac{\hat{q} L_2^2}{2k_2}, \quad q_{12} = q_0 - \left[q_{01} \left(1 + \frac{a L_1^2}{2} \right) \right] \Rightarrow A$$

$$\Delta T_3 = -\frac{q_{23} L_3}{k_3} + \frac{q_{03}}{k_3} \left(\frac{L_3^2}{2} + \frac{a L_3^3}{6} \right), \quad q_{23} = q_{12} + \hat{q} L_2 = q_0 - A + \hat{q} L_2$$

$$R_{\text{tot}} = \frac{L_1}{k_1} + \frac{L_2}{k_2} + \frac{L_3}{k_3}$$

$$\therefore C = \frac{q_{01}}{k_1} \left(\frac{L_1^2}{2} + \frac{a L_1^3}{6} \right) + A \left(\frac{L_2}{k_2} + \frac{L_3}{k_3} \right) - \hat{q} \left(\frac{L_2^2}{2k_2} + \frac{L_2 L_3}{k_3} \right) + \frac{q_{03}}{k_3} \left(\frac{L_3^2}{2} + \frac{a L_3^3}{6} \right)$$

$$\text{Thus, } T_{s,2} - T_{s,1} = -q_0 R_{\text{tot}} + C$$

$$q_0 = \frac{(T_{s,1} - T_{s,2}) + C}{R_{\text{tot}}}$$

$$q_{12} = q_0 - A$$

$$q_{23} = q_{12} + \hat{q} L_2 \quad \checkmark$$

Temperature profiles (R₁) $T_1(x) = T_{s,1} - \frac{q_0}{k_1} x + \frac{q_{01}}{k_1} \left(\frac{x^2}{2} + \frac{a x^3}{6} \right)$

(R₂) $\Rightarrow T_2(x) = T_1(L_1) - \frac{q_{12}}{k_2} (x - L_1) - \frac{\hat{q}}{2k_2} (x - L_1)^2$

$$T_1(L_1) = T_{s,1} - \frac{q_{01} L_1}{k_1} + \frac{q_{01}}{k_1} \left(\frac{L_1^2}{2} + \frac{a L_1^3}{6} \right)$$

(R₃) $\Rightarrow T_3(x) = T_2(L_1 + L_2) - \frac{q_{23}}{k_3} x + \frac{q_{03}}{k_3} \left(\frac{x^2}{2} + \frac{a x^3}{6} \right)$

$$T_2(L_1 + L_2) = T_1(L_1) - \frac{q_{12} L_2}{k_2} - \frac{\hat{q} L_2^2}{2k_2}$$

$$\therefore q''(x) = \begin{cases} \text{For (R}_1\text{)} \rightarrow q_0 - q_{01} \left(x + \frac{a x^2}{2} \right) \\ \text{For (R}_2\text{)} \rightarrow q_{12} + \hat{q} (x - L_1) \\ \text{For (R}_3\text{)} \rightarrow q_{23} - q_{03} \left(x + \frac{a x^2}{2} \right) \end{cases}$$